



Hepatic and Biliary Fascioliasis

Microbiology GIT Block · MEDucation by Mattin Majid

BEFORE YOU READ — ANSWER FROM MEMORY

1. What are the two intermediate hosts in the *F. hepatica* life cycle, and which is first vs second?
2. How does the acute (migratory) phase of fascioliasis differ from the chronic (biliary) phase?
3. What is halzoun, and how is it acquired?

Guessing wrong before reading still improves retention — attempt all three, then check as you go.

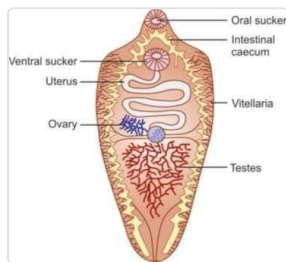
1 *Fasciola hepatica* — the sheep liver fluke

- Common name: **sheep liver fluke**; the **largest and most common liver fluke** found in humans, but its primary host is the **sheep** (to a lesser extent, cattle)
- **Worldwide**, mainly in sheep-rearing areas; causes the economically important disease "**liver rot**" in sheep
- **Habitat**: resides in the **liver and biliary passages** of the definitive host

Stage	Feature
Adult worm	Large, leaf-shaped , fleshy fluke, 30 mm long, 15 mm broad , grey/brown; conical anterior projection with an oral sucker , rounded posteriorly; lives in the biliary tract for ~5 years in sheep, ~10 years in humans ; hermaphrodite
Egg	Large, ovoid, operculated, bile-stained (yellow-brown) , about 140 × 80 µm



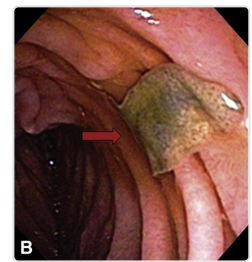
Adult fluke — leaf-shaped, fleshy.



Internal anatomy: oral/ventral suckers, intestinal caecum, uterus, ovary, vitellaria, testes.



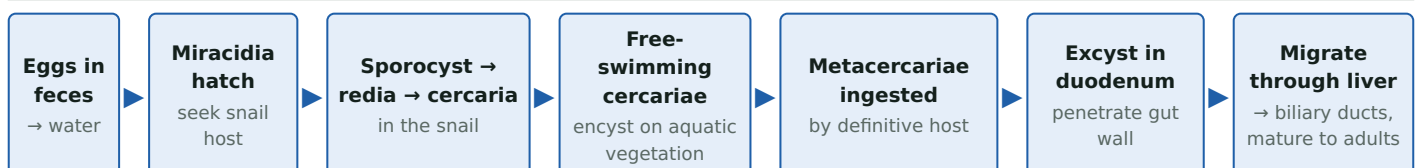
Operculated, bile-stained egg.



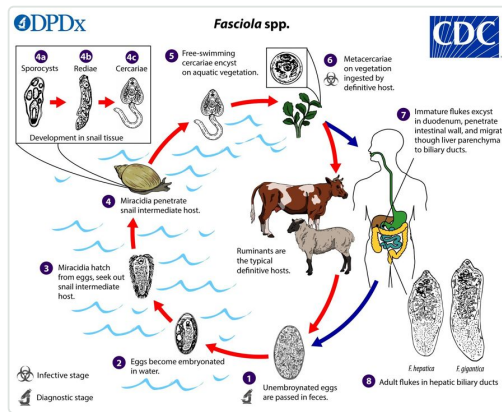
Fluke seen on endoscopy in the biliary tree.

2 Life cycle — one definitive host, two intermediate hosts

Role	Organism
Definitive host	Sheep, goat, cattle, and human
1st intermediate host	Snails
2nd intermediate host	Aquatic plants (e.g. watercress)
Infective stage	Metacercariae
Mode of infection	Ingestion of metacercariae encysted on aquatic vegetation



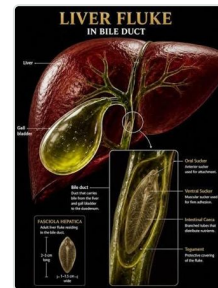
Immature flukes penetrate the intestinal wall and migrate through the liver parenchyma before reaching the bile ducts.



Full life cycle (DPDx/CDC): snail → aquatic vegetation → ingestion → hepatic migration → biliary ducts.



Labelled morphology of the adult fluke.



Fluke in the gallbladder and bile duct.

3 Clinical manifestations — two phases

Phase	Timing / mechanism	Presentation
Acute (migratory)	Larva migrates through and traverses liver tissue , causing mechanical/parenchymal injury; symptoms subside as parasites reach their final destination	Fever, right upper quadrant pain, eosinophilia, tender hepatomegaly
Chronic (biliary)	Flukes reside in the bile ducts and mature into adults over ~3-4 months	Biliary obstruction, biliary cirrhosis, obstructive jaundice, cholelithiasis, anemia

Halzoun: ingestion of raw infected sheep liver → adult worms attach to the pharyngeal mucosa → edematous pharyngeal congestion → dyspnea, acute dysphagia, deafness, rarely asphyxiation ("halzoun" = suffocation).

4 Diagnosis, treatment, and management

	Detail
Stool microscopy	Best method: demonstration of eggs in feces or aspirated duodenal bile
Serodiagnosis	ELISA becomes positive within 2 weeks of infection
Imaging	USG, CT scan may help
Treatment of choice	Oral triclabendazole
Alternative drug	Bithionol
Biliary obstruction	Manual extraction of adult flukes via ERCP (endoscopic retrograde cholangiopancreatography) if physical blockage occurs

5 Prevention

- Health education
- Prevent pollution of water courses with **sheep, cattle, and human feces**
- Avoid eating **raw watercress or other freshwater aquatic plants** grown in endemic areas
- Avoid eating **raw liver** of an infected animal (prevents halzoun)
- Proper **disinfection of watercress** and other water vegetation before consumption
- Control the **intermediate host (snail)**

P Must-remember pearls

- **Two intermediate hosts:** snail (1st) → aquatic vegetation (2nd); **infective stage = metacercariae**, acquired by **eating watercress**.
- **Acute phase** = hepatic migration → fever, RUQ pain, eosinophilia, hepatomegaly. **Chronic phase** = biliary residence (3–4 months to mature) → obstruction, cirrhosis, jaundice, cholelithiasis.
- **Halzoun** = eating raw infected liver → pharyngeal attachment → dyspnea/dysphagia/deafness, rarely asphyxiation.
- Diagnosis: **eggs in stool or duodenal bile** is best; ELISA positive by 2 weeks.
- Treatment: **triclabendazole** (first line), **bithionol** (alternative); ERCP for biliary obstruction.

Q Test yourself

1. A shepherd who eats raw watercress from a sheep-grazing area develops fever, right upper quadrant pain, eosinophilia, and tender hepatomegaly. Which phase and mechanism explain this?

- A. Acute phase — mechanical damage from larval migration through liver parenchyma
B. Chronic phase — biliary obstruction by adult worms
C. Halzoun — pharyngeal attachment of adult flukes
D. Chronic phase — obstructive jaundice from cholelithiasis

2. Which stage of Fasciola hepatica is infective for the definitive host, and how is it acquired?

- A. Metacercariae, by ingesting encysted forms on aquatic vegetation
B. Cercariae, by skin penetration in water
C. Miracidia, by ingesting snails
D. Rhabditiform larvae, by ingesting contaminated soil

3. A man who ate raw sheep liver develops dyspnea, dysphagia and pharyngeal swelling. Diagnosis?

- A. Halzoun
B. Acute hepatic fascioliasis
C. Biliary cirrhosis
D. Cholelithiasis

4. What is the treatment of choice for fascioliasis?

- A. Oral triclabendazole
B. Metronidazole
C. Praziquantel
D. Mebendazole

5. Which specimen best demonstrates Fasciola hepatica eggs for diagnosis?

- A. Feces or aspirated duodenal bile
B. Peripheral blood smear
C. Sputum
D. Urine sediment

ANSWERS — COVER UNTIL YOU'VE COMMITTED

- 1 — A** — The acute phase reflects larval migration causing parenchymal injury; symptoms subside once the parasite reaches the bile ducts.
2 — A — Humans and livestock are infected by ingesting metacercariae encysted on aquatic plants such as watercress.
3 — A — Halzoun follows ingestion of raw liver harbouring adult flukes that attach to the pharyngeal mucosa.
4 — A — Triclabendazole is the drug of choice; bithionol is the alternative.
5 — A — Eggs in stool or aspirated bile are the best diagnostic method.

SPACED REVIEW

☐ Day 1

☐ Day 3

☐ Day 7

☐ Day 21

☐ Pre-exam